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Scientific Software Developer

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Dear D-Wave Hiring Team,

I am applying for the Senior Scientific Software Developer position in Burnaby. I am a scientific software developer with over 20 years of hands-on experience in C/C++, numerical mathematics, high-performance computing, and software architecture. At Ulm University, I design, implement, and maintain numerical and systems software together with scientists and engineers. Designing, implementing, debugging, and optimizing scientific software is what I enjoy most and what I want to focus on professionally.

Much of my work has gone into numerical libraries. FLENS, the C++ framework I developed, maps high-level matrix abstractions directly to optimized BLAS and LAPACK kernels without abstraction overhead; my BLAS library ulmBLAS separates portable blocking algorithms from small, architecture-specific micro-kernels and reaches performance close to vendor libraries. Maintaining both over many years has taught me to treat architecture, testing, benchmarking, and documentation as part of the engineering work.

With the Institute of Quantum Technologies at the German Aerospace Center (DLR), I am developing HeraldRT, a SystemVerilog design for FPGA-based real-time control of quantum-memory experiments. It has been simulated, synthesized, placed, and routed; testing on physical hardware is the next step. The work requires translating the physicists' requirements into deterministic hardware behavior, meeting timing constraints, and defining clean hardware/software interfaces; it builds directly on my background in processor architecture, FPGA design, and low-level programming.

As managing director of the Ulm Center for Scientific Computing, I translated problems of industrial and research partners into mathematical models and working software, often implementing the complete application myself. I am used to taking ownership of a problem: understanding the requirements, investigating failures until the cause is clear, implementing a solution, and validating the result.

What attracts me to this position is that it combines hands-on scientific software engineering, from calibration to hardware interfaces and system testing, with direct collaboration with the physicists and hardware engineers who develop D-Wave's quantum processors. My quantum-related experience so far concerns quantum-memory experiments rather than superconducting processors, and I look forward to learning D-Wave's technology from the people who build it. I am willing to relocate to the Vancouver/Burnaby area.

Thank you for your consideration.

Sincerely,

Michael Lehn